

Rates and Ratios Test Preparation

Simplifying Ratios

a \$4:\$12

= _____

b 30 cents:\$6

= _____

c 10 hours:1 day

= _____

d 400 g:1 kg

= _____

e 50 cents:90 cents

= _____

f 3 days:3 weeks

= _____

g 12 cm:1 m

= _____

h 20 sec:1 min

= _____

i 2 kg:800 g

= _____

j 400 mL:1 L

= _____

k 3 km:600 m

= _____

l 8 cm:60 mm

= _____

10 2.0 : 0.5 and 8 : 2

11 2.5 : 0.5 and 10 : 2

12 6 : 4 and 1.8 : 1.2

Applying Ratios

- a** The ratio of boys to girls is 2:3. If there are 12 boys, how many girls are there?

- b** Two girls aged 12 years and 15 years divide \$72 in the ratio of their ages. How much does each girl receive?

- c** Find the ratio of the areas of two squares whose sides are 3 cm and 5 cm, respectively.

- d** A piece of rope is cut into three lengths in the ratio 2:3:5. If the smallest length is 6 m, find the length of the original rope.

- c** Two brothers are to share an inheritance of half a million dollars in the ratio 12:13. How much is the smaller share?

- d** The ratio of adults to children on a train trip is 3:2. If the train carries 800 passengers, find the number of adults and children on the train.

Rates

QUESTION 1 Complete the following sentences.

- a 320 km in 5 hours is a rate of _____ per hour.
- b 48 books bought for \$360 is at the rate of _____ per book.
- c If 900 litres of water flows through a tap in 2 hours, it is a rate of _____ per minute.
- d Richard works for 10 hours and is paid \$145. His rate of pay is _____ per hour.

QUESTION 2 Complete the equivalent rates.

- a 90 km/h = _____ km/min
- b 10 L/h = _____ L/day
- c 8 m/min = _____ m/h
- d \$3/min = _____ \$/h
- e 20 mL/min = _____ mL/h
- f 30°/min = _____ °/second

QUESTION 3

- a Michael drives 180 km in 3 hours. Find his average speed. _____

- b Change 120 km/h into km/min.

- c Change 180 km/h into m/s.

- d John delivered 540 bottles of milk every morning between 5:00 am and 9:00 am. Find his hourly rate of delivery.

- e A car travels at the speed of 25 m/s. How many kilometres does it travel in 1 hour?

QUESTION 4

- a A car travels 600 km on a journey and covers this distance in 6 hours and 15 minutes. Calculate the average speed per hour.

- b A tree grows to a height of 32.4 metres over a period of $8\frac{1}{2}$ years. What is the average annual growth rate in metres per year?

- c The distance between two towns is 90 km. A bus driver averages 30 km/h in heavy traffic in one direction and 45 km/h on the return trip. Calculate the total time taken and the average speed for the total journey.

